ECE 2414: Digital Communications
LAB 1 Analysis of Sampling Theorem Using MATLAB

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1 Objective

To analyze and verify the Sampling Theorem, reconstruct the original signal from sampled data, and perform quantization using MATLAB.

2 Introduction

The Sampling Theorem states that a continuous-time signal can be fully reconstructed from its samples if it is band-limited and the sampling frequency is at least twice the maximum frequency of the signal (Nyquist rate). This experiment investigates the sampling process, analyzes its frequency-domain representation, and demonstrates reconstruction using a low-pass filter (LPF).

3 Acknowledgment

The MATLAB code utilized in this experiment is credited to Copyright @ Dr. Sudip Mandal, Assistant Professor, Jalpaiguri Government Engineering College, ECE Department, West Bengal, India, PIN-735102. The code is part of the Digital Communication Lab resources developed by Dr. Mandal. I sincerely thank Dr. Mandal for making these educational resources accessible, which significantly contributed to the understanding and execution of this lab experiment.

4 Experiment1: Analysis of Sampling Theorem

Procedure

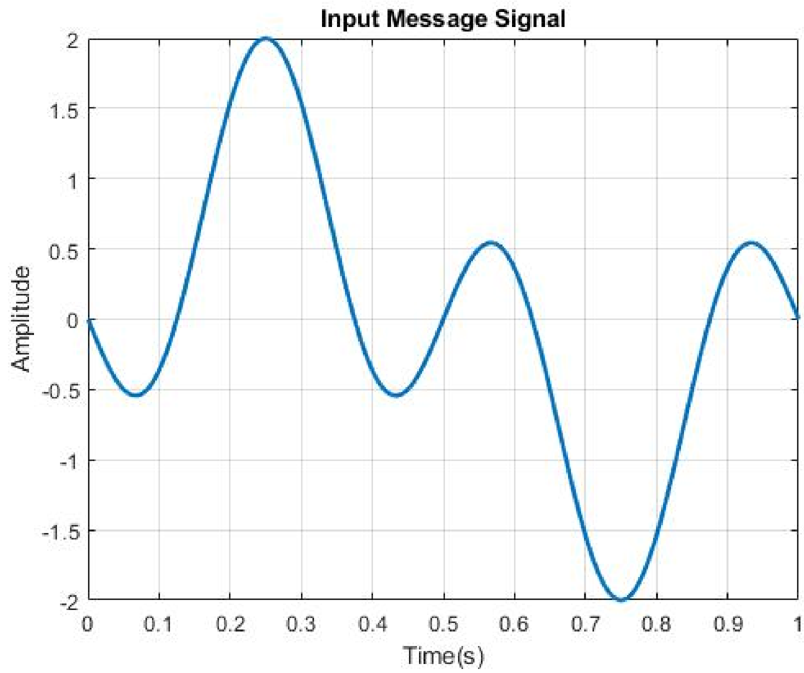

Figure 1: Input Message Signal

1. A message signal with sinusoidal components at 1 Hz and 3 Hz was generated.
2.The signal was plotted in the time domain.
3. The frequency spectrum was computed using the Fast Fourier Transform (FFT).
4. The signal was sampled at a rate of 50 Hz.
5. The sampled signal and its frequency spectrum were plotted.

The resulting plot displays a time-domain representation of the message signal, which is the sum of the two sinusoidal components. The signal exhibits a periodic pattern due to the combination of the 1 Hz and 3 Hz frequencies. The lower frequency (1 Hz) component dominates the overall shape of the signal, while the higher frequency (3 Hz) component introduces additional oscillations within each cycle of the 1 Hz signal. The subtraction of the two sinusoids creates a waveform with constructive and destructive interference patterns, leading to variations in amplitude over time. The fine time resolution (sampling interval of 0.002 seconds) ensures a smooth and accurate plot of the signal. Observing the waveform provides insight into how the superposition of different frequency components forms the overall signal. This visualization can be used as a basis for analyzing the signal further, such as studying its frequency spectrum or

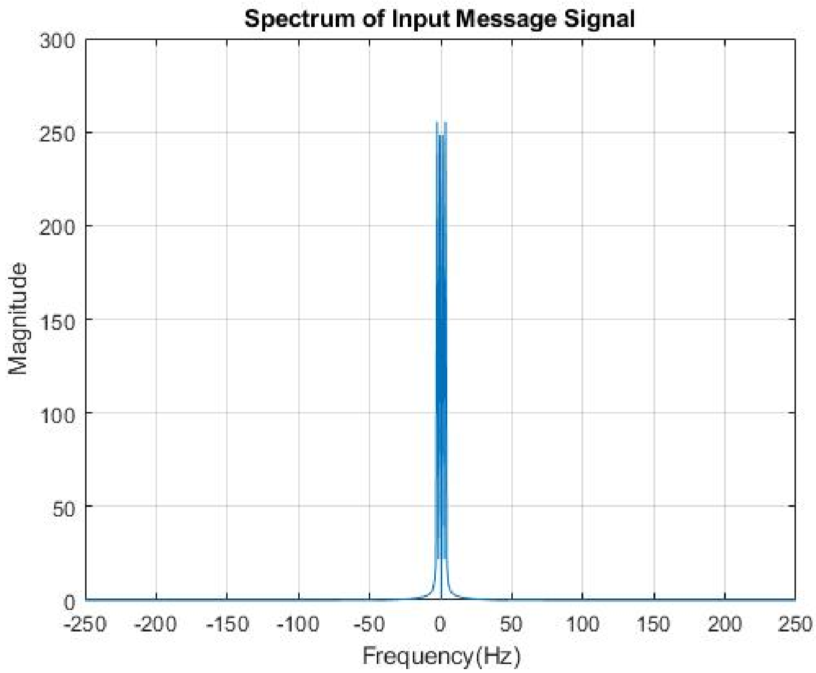

Figure 2: Spectrum of Input Signal

sampling properties in subsequent steps.

The plot shows the magnitude of the frequency components of the input message signal x. Peaks appear at frequencies corresponding to the sinusoidal components present in x (e.g., 1 Hz and 3 Hz if the signal has these components). The spectrum is symmetric about the origin (due to the real nature of the input signal), with positive and negative frequency components represented on the frequency axis. If the sampling rate (determined by td) satisfies the Nyquist criterion (sampling rate ¿2 $\times$ highest frequency in the signal), the peaks in the spectrum will appear correctly. If the sampling rate is too low (aliasing occurs), the peaks may overlap, leading to distorted frequency components. Using a zero-padded FFT length ensures higher resolution in the frequency domain, making the plot smoother and the frequency components easier to identify.

The stem plot displays the sampled signal in the time domain. Each point represents the amplitude of the signal at the corresponding sampling time n. The plotted signal shows a combination of a 1 Hz sinusoid and an inverted 3 Hz sinusoid. Since the sampling frequency (50 Hz) is significantly higher than

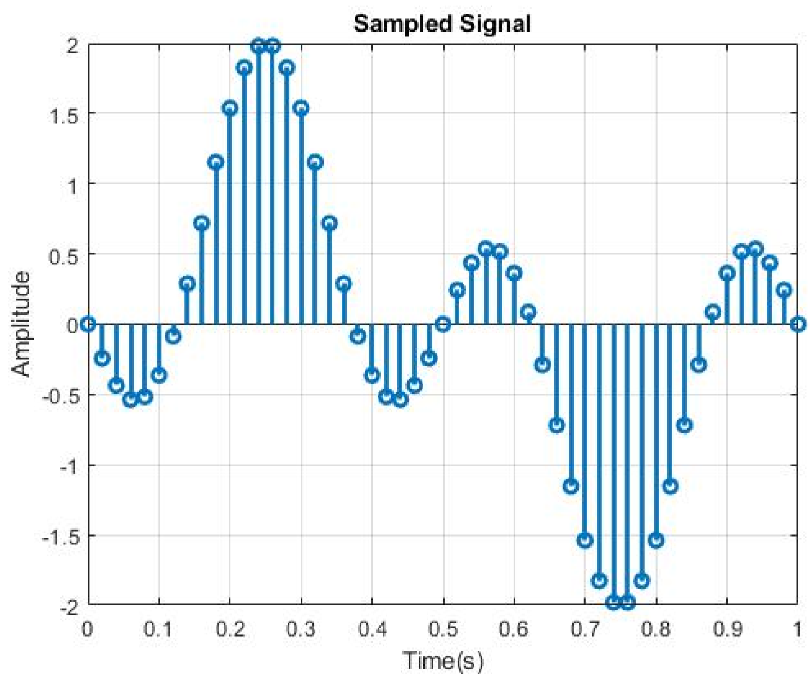

Figure 3: sampled signal

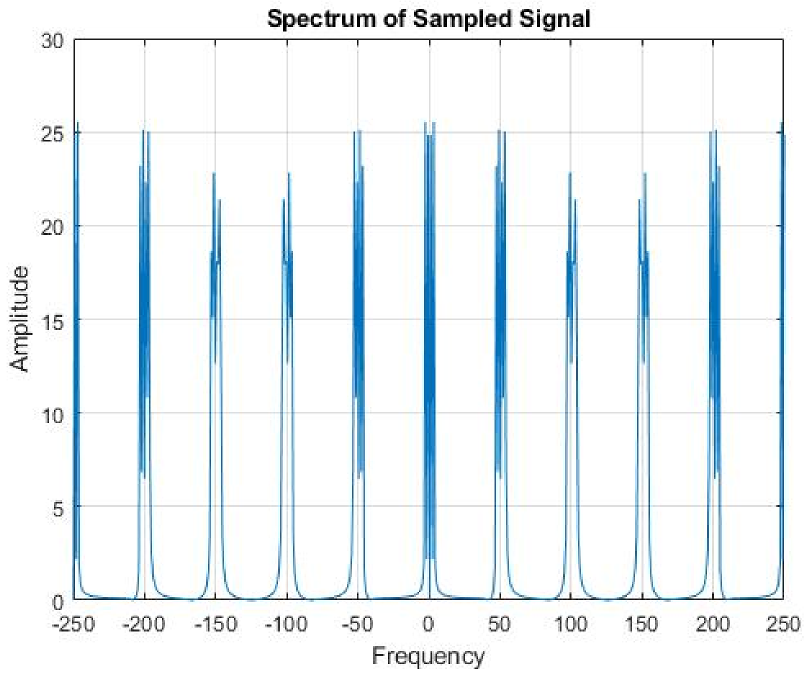

Figure 4: spectrum of sampled signal

twice the highest frequency component (6 Hz), aliasing will not occur. The plot confirms the preservation of the signal's characteristics after sampling due to the adherence to the Nyquist criterion. The 1 Hz component appears as a slower oscillation, while the 3 Hz component appears as faster oscillations superimposed on the 1 Hz wave. The use of a discrete stem plot visually emphasizes the discrete nature of the sampled signal, contrasting with the continuous-time representation.

The original frequency components of the signal appear as distinct peaks. If the signal was sampled at a rate below the Nyquist rate, aliasing may occur, resulting in additional frequency components in the spectrum. Up sampling increases the number of points used in the FFT computation, enhancing the resolution of the frequency spectrum. This allows for a clearer distinction between closely spaced frequency components. Since the FFT includes both positive and negative frequencies, the spectrum will be symmetric around 0 Hz. By observing the spectrum, it can be verified whether the sampling rate was sufficient to avoid aliasing. If aliasing occurred, the reconstructed signal will show distorted or misplaced frequency components.

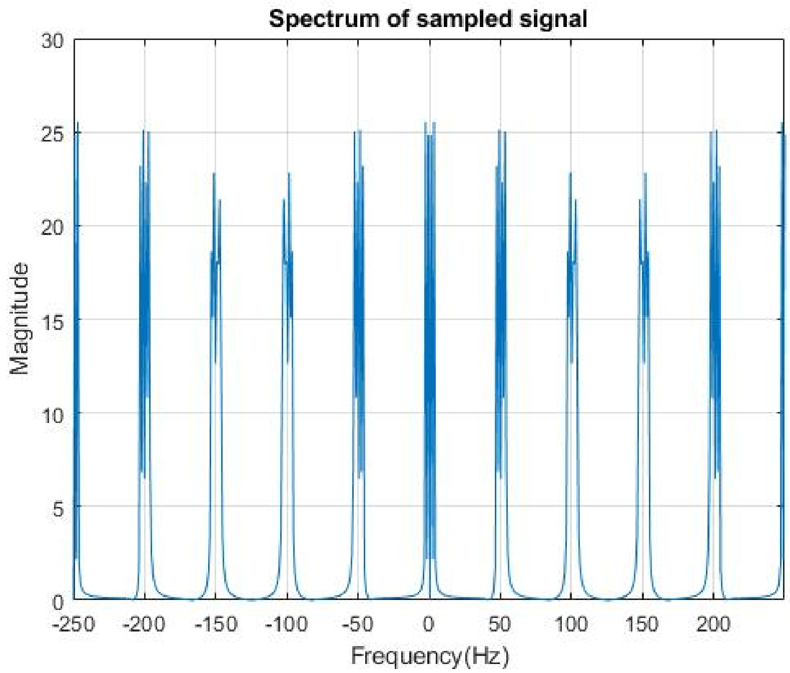

Figure 5: spectrum of sampled signal

5 Experiment2: Reconstruction from Sampled Signal

Procedure

1. The sampled signal was upsampled to match the original signal's length.
2. A low-pass filter was designed with a cutoff frequency of 10 Hz.
3. The upsampled signal was passed through the low-pass filter to reconstruct the original signal.
4. The reconstructed signal was compared with the original message signal in the time domain.

The FFT of the upsampled signal (xsmu) is plotted. The spectrum shows the peaks corresponding to the original signal components (1 Hz and 3 Hz), depending on whether aliasing occurred during downsampling. It also shows the artifacts from the upsampling process, including additional spectral components due to the zero-padding.

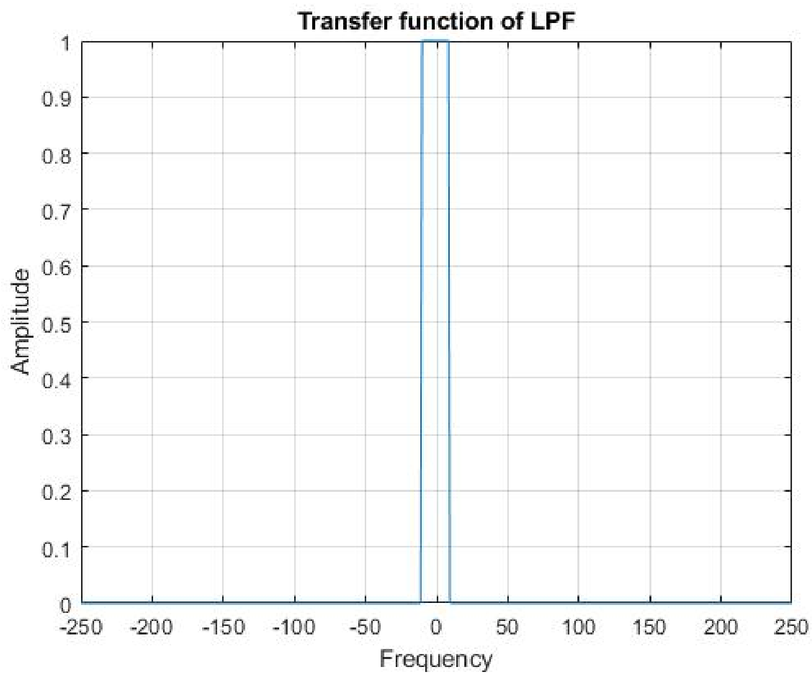

Figure 6: transfer function lpf

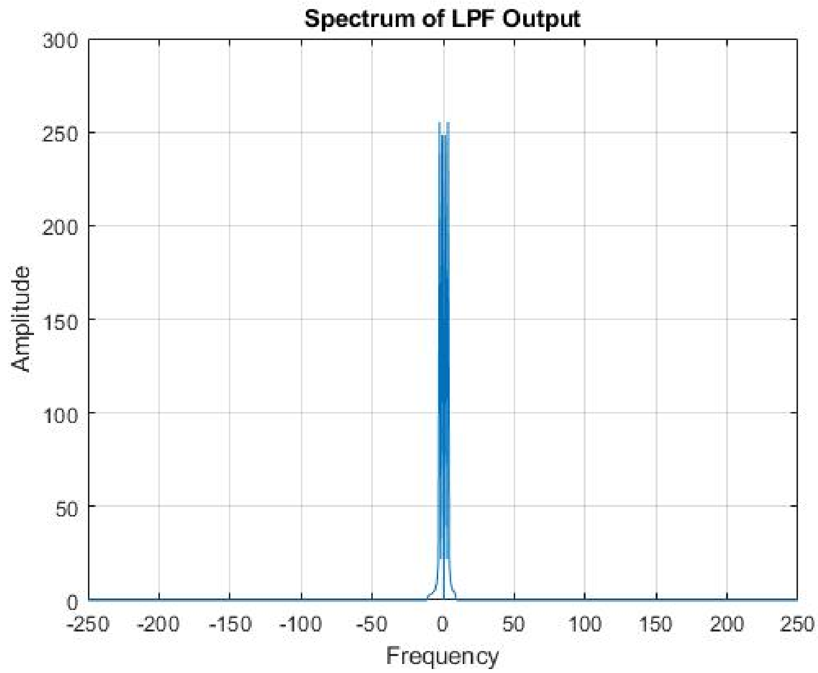

Figure 7: transfer function of lpf output

The resulting plot shows the transfer function of a low-pass filter with a passband centered around the middle frequency (at Lffu/2). The filter passes frequencies within a bandwidth of BW = 10 around this middle frequency, and all other frequencies will be blocked (amplitude=0). The plot looks like a rectangular shape, with the amplitude equal to 1 for frequencies within the passband and 0 for frequencies outside the passband.

The output is a frequency-domain plot (spectrum) of the signal x_recv, showing how the low-pass filter has affected its amplitude across different frequencies. The low-pass filter will attenuate higher frequencies and pass lower frequencies, and the magnitude of the spectrum will be adjusted by the Nfactor scaling.

The plot shows two signals: the original signal x (in red) and the reconstructed signal x_recv2 (in blue dashed line). The FFT and IFFT operations were performed correctly and without significant noise or distortion, and hence the reconstructed signal closely matches the original signal. The output visually shows the comparison between the original and reconstructed signal, where ideally, the two signals should be almost identical, demonstrating that the signal processing (FFT and IFFT) was successful.

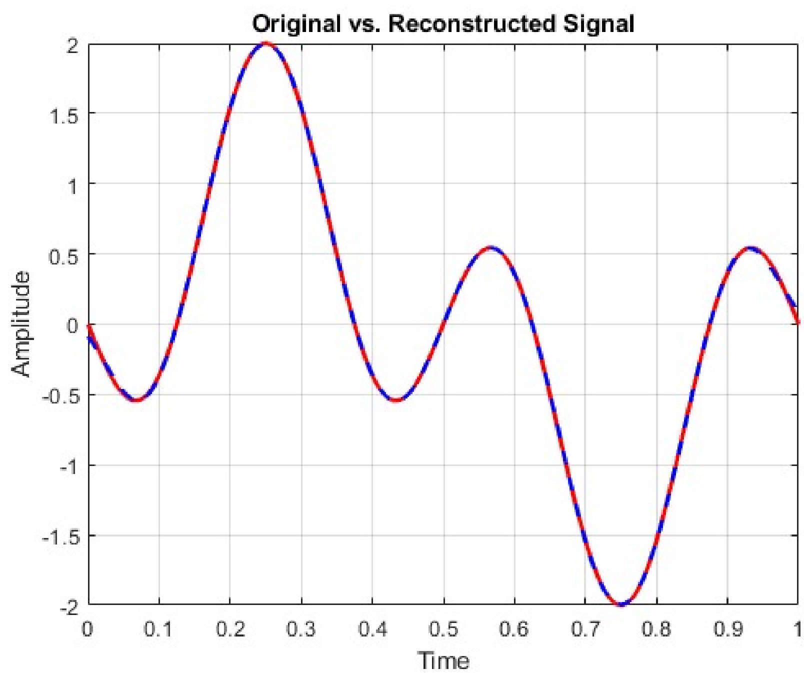

Figure 8: Original Vs Reconstructed Signal

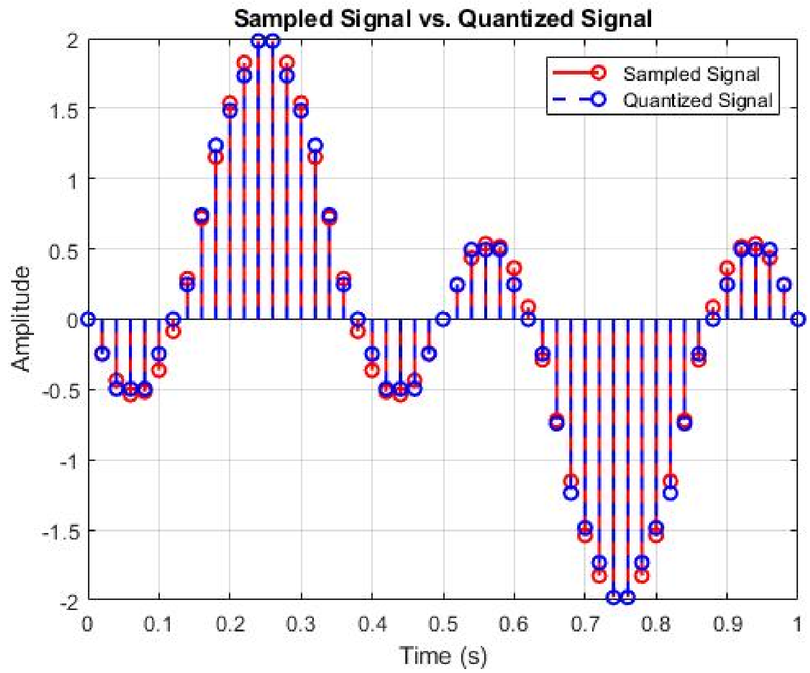

Figure 9: Sampled Signal vs Quantized Signal

6 Quantization

Procedure

1. Quantization levels (e.g., 8, 16, or 32 levels) are defined.
2. The sampled signal is quantized by rounding to the nearest discrete level.
3. Quantization error is calculated as the difference between the original and quantized signals.
4. MATLAB and Python implementations of quantization are provided.

7 Discussion Questions

1.Theory

Why is the Nyquist rate important for the sampling process?
The Nyquist rate ensures the sampled signal retains all the information of the original signal, avoiding aliasing.

2.Spectrum Analysis

Describe the frequency spectrum of the sampled signal. How does it change with different sampling rates?
. When the sampling rate is at or above the Nyquist rate, the spectrum replicates cleanly without overlap.
. At lower sampling rates, spectrum overlap occurs, causing aliasing.

3.Reconstruction

How does the low-pass filter affect reconstruction?
. The LPF removes high-frequency components beyond the Nyquist frequency, allowing recovery of the original signal.
. If the LPF bandwidth is reduced, it can attenuate components of the original signal. If increased, noise or aliasing artifacts may pass through.

4.Aliasing

What is aliasing, and how does it appear in the spectrum of the sampled signal? How can you avoid aliasing in a practical sampling system?
. Aliasing occurs when the sampling rate is below the Nyquist rate, causing spectrum overlap. It appears as incorrect frequency components in the spectrum. Using an anti-aliasing filter and ensuring a sufficient sampling rate avoids aliasing.

5.Effects of Undersampling

If the sampling rate is below the Nyquist rate, how would this affect the reconstruction?
. Undersampling distorts the reconstructed signal. In the time domain, the signal may have incorrect oscillations, and in the frequency domain, aliasing causes spectral overlap.

6.Practical Sampling Rates

Why might we choose a sampling rate higher than the minimum Nyquist rate?
. Higher sampling rates provide robustness against aliasing and ease of filter design while preserving signal integrity.

8 Extension Task: Additional Questions

1.Quantization Error:

How does the quantization error change with the number of quantization levels? Describe the relationship between quantization levels and signal quality.
. Quantization error decreases with increasing quantization levels, improving signal quality.

2.Signal-to-Noise Ratio (SNR)

$$SNR(dB) = 6 \cdot 02log2(L) + 1 \cdot 76$$

$$L = 8 : SNR = 6.02log2(8) + 1.76 = 6.023 + 1.76 = 19.82dB$$

$$L = 16 : SNR = 6.02log2(16) + 1.76 = 6.023 + 1.76 = 25.84dB$$

$$L = 32 : SNR = 6.02log2(32) + 1.76 = 6.023 + 1.76 = 31.86dB$$

3.Bitrate Calculation

$$Bitrate\ (bits\ per\ second) = fslog2(L)$$

$$Where :$$

$$fs = 50Hz(samplingrate)$$

$$L : Numberofquantizationlevels.$$

$$L = 8 : Bitrate = 50log2(8) = 50 \cdot 3 = 150bps$$

$$L = 16 : Bitrate = 50log2(16) = 50 \cdot 4 = 200bps$$

$$L = 32 : Bitrate = 50log2(32) = 50 \cdot 5 = 250bps$$

Impact of Increasing Sampling Rate or Quantization Levels:
1. Increasing Sampling Rate:
Higher fs improves signal fidelity, as it captures more details of the original signal. However, it increases the *bitrate*, requiring more bandwidth and storage.
2. Increasing Quantization Levels (b):
Higher b reduces quantization error and improves SNR, leading to better signal quality. It also increases the *bitrate*, demanding more resources for data transmission and storage.

9 Conclusion

This experiment successfully demonstrated the principles of sampling and quantization, highlighting the importance of the Nyquist rate and quantization levels for effective digital communication.

10 References

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